Why the direct-minimization embedding state is an excited eigenstate

(ghost Gutzwiller for the t_c-Cederbaum model, $B_c = 3$)

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This note explains a numerical observation made when the solution of the direct minimization (`not_imp_gga_cederbaum.py`) was used as the starting point of the embedding fixed-point loop (`gga_cederbaum.py`) for the t_c-Cederbaum model of `cederbaum.tex`: the core embedding state Φ_c of the direct minimization is an *excited* eigenstate of the embedding Hamiltonian, which the loop (that takes the lowest state) cannot follow. Notation is that of `cederbaum.tex`.

1 Statement

"Excited" does *not* refer to the physical system: the direct-minimization point is the lowest-energy solution of the constrained problem (its energy agrees with the exact ground-state energy to $\sim 10^{-7}$). It refers to the embedding Hamiltonian $H^c_{\rm imp}$ built from the multipliers recovered at that point: Φ_c is an exact eigenstate of $H^c_{\rm imp}$, but not the lowest one.

2 The two procedures and what each requires

Both stationarize the same Lagrangian

$$\mathcal{L} = E + \sum_I 2\,\mathrm{Tr}\big[\lambda_I \big({}^I\Delta(\Phi_I) - \rho^{II}\big)\big] - \sum_I E^c_I \big(\langle\Phi_I|\Phi_I\rangle - 1\big), \tag{1}$$

with E of Eq. (E) of `cederbaum.tex` and multipliers λ_I (density constraint) and E^c_I (normalization).

Direct minimization minimizes E over Φ_I and ψ_{qp} subject to the constraints. At a constrained minimum, $\mathcal{L}$ is stationary in Φ_I, i.e.

$$H^I_{\rm imp}\,\Phi_I = E^c_I\,\Phi_I, \tag{2}$$

where, for the core,

$$H^c_{\rm imp} = H^{loc}_c + \sum_{a\sigma} h_a \big(d^\dagger_{a\sigma} c_\sigma + \mathrm{h.c.}\big) + \sum_{ab\sigma} \lambda^c_{ab}\, d^\dagger_{a\sigma} d_{b\sigma} + n_c \Big[\alpha + 2\mathrm{Tr}\Xi - \sum_{ab\sigma} \Xi_{ab}\, d^\dagger_{a\sigma} d_{b\sigma}\Big], \qquad \lambda^c = \Lambda - \lambda, \tag{3}$$

(with h_a, Λ, α, Ξ the derivatives of E given in Sec. Direct minimization of `cederbaum.tex`). Equation (2) says that Φ_I is *an eigenstate* of $H^I_{\rm imp}$; it says nothing about which one.

The impurity loop *assumes* that Φ_I is the *lowest* eigenstate of $H^I_{\rm imp}$ at the current multipliers (it diagonalizes $H^I_{\rm imp}$ and takes the ground state). This is the saddle-point reading of $\mathcal{L}$: minimize over Φ_I at fixed multipliers, then adjust the multipliers to satisfy the constraints. It is equivalent to the constrained minimum when the problem has no duality gap (e.g. if it were convex).

3 Numerical check

At $t_c = 0.5$, $B_c = 3$ ($\Delta = 1$, $E_c = 10$), for the two couplings $V/g = 1/\sqrt{2} = 0.707$ and $\sqrt{2} = 1.414$ ($\rho/\Omega = V/(\sqrt{2}g) = 0.5$ and 1.0, the ratios of Kockläuner and Golze; see `cederbaum.tex`), with the multipliers recovered from the direct-minimization point (shift $\varepsilon = 10^{-9}$; `excited_check_golze.py`):

	$V/g = 0.707$	$V/g = 1.414$
overlap^2 of Φ_c with the selected eigenstate of $H^c_{\rm imp}$	1.000000	1.000000
index of that eigenstate (0 = lowest; $H^c_{\rm imp}$ has 36 states)	3	5
$\langle\Phi_c\|H^c_{\rm imp}\|\Phi_c\rangle$	-22.04	-22.04
lowest eigenvalue of $H^c_{\rm imp}$	-24.32	-30.19
index of Φ_o, Φ_v in their $H_{\rm imp}$ (4 states each)	0, 0	0, 0
bath levels of λ^c_c (eigenvalues of the 3×3 matrix)	-2.14, -1.73, $+1.73$	-5.07, -3.00, $+3.00$
eigenvalues of ${}^c\Delta$	6.0×10^{-6}, 0.9979, 1.0000	2.5×10^{-5}, 0.9978, 1.0000

That is: Φ_c is an exact eigenstate of $H^c_{\rm imp}$, as Eq. (2) requires; it is not the lowest one (the fourth and the sixth of the 36 eigenstates, with the lowest eigenvalue 2.3 and 8.1 below it), whereas Φ_o and Φ_v are ground states. The dimensions are those of the half-filled sector $N_\uparrow = N_\downarrow = (1+B)/2$ of the fragment (physical orbital plus B bath orbitals per spin): for the core, 4 orbitals per spin with 2 electrons of each spin, $\binom{4}{2}^2 = 36$; for o and v, 2 orbitals per spin with 1 electron of each spin, $\binom{2}{1}^2 = 4$.

How the size of $H^I_{\rm imp}$ is obtained. The embedding problem of fragment I has the physical orbital c and B_I bath orbitals d_a per spin (the bath orbitals include the ghosts: $B_c = 3$ is one bath level plus two ghosts), i.e. $M = B_I + 1$ orbitals per spin and $2M$ spin orbitals. The full Fock space has dimension 2^{2M}. The Hamiltonian $H^I_{\rm imp}$ conserves $N_\uparrow$ and $N_\downarrow$, and the code (`Fragment` in `gga_cederbaum.py`) projects it on the sector with $N_\uparrow = N_\downarrow = M/2$ (half filling, which requires B_I odd). Spin up and spin down are independent in the counting, so

$$\dim H^I_{\rm imp} = \binom{M}{M/2}^2, \qquad M = B_I + 1. \tag{4}$$

The operators are built in the full Fock space with sparse Jordan–Wigner matrices and then restricted to this sector (stored dense), and the dimension printed above is the shape of the projected `Hloc`. The values are

B_I	M	full Fock space 2^{2M}	sector $N_\uparrow = N_\downarrow = M/2$
1	2	16	4
3	4	256	36
5	6	4096	400

so $B_c = 3$ and $B_c = 5$ give 36 and 400 states for the core, and 4 for each of o and v ($B_o = B_v = 1$). The number of states grows much faster than the number of ghosts, but the solution only needs

the one eigenstate Φ_I that satisfies the constraint, not all of them. The index of the excited state is not a property of the model alone: it depends on the multipliers recovered at the point, and these contain one direction that the stationarity conditions leave undetermined (Sec. *Absence of satellites* of `cederbaum.tex`).

4 Why it can be excited

1. **Stationarity is not minimality.** Stationarity of $\mathcal{L}$ in Φ gives Eq. (2) for *some* eigenstate. It is the *constraint*, not the energy of H_{imp}, that selects which one.

2. **The constraint fixes the bath occupations.** ${}^c\Delta = \mathbb{I} - \Delta^{bb}$ must equal ρ^{cc}, the local density matrix of the quasiparticle state. With one bath electron per spin, this fixes *which* bath orbital is filled: the one that corresponds to the empty quasiparticle orbital. The multipliers λ_I are then adjusted so that this state is stationary.

3. **That need not be the lowest state at the recovered multipliers.** At these multipliers the lowest state of H^c_{imp} would put the bath electron in the lowest bath level (-2.14 and -5.07 at the two couplings) (and, with it, change the core occupation), which contradicts the constraint satisfied by Φ_c. The state that satisfies the constraint is a different, higher eigenstate.

4. **Non-convexity.** The constraint is quadratic in Φ and the energy functional is nonlinear in the quantities extracted from Φ (through $R_I = S^{-1}D$ and $\bar{\Theta}$), so the problem is not convex and a duality gap is possible: the minimum of $\mathcal{L}$ over Φ at the optimal multipliers is not the constrained minimum. This is why the saddle-point assumption of the loop can fail.

Points 1 and 2 follow directly from the equations; points 3 and 4 are my reading of the numerical result, not a proof.

5 Consequences for the impurity loop

Lowest-state selection. Started from the full direct-minimization state (`state0`), one loop iteration already moves away: the residual $\|\rho^{II} - (1 - \Delta^{bb}_I)\|$ is 1.41 at both couplings (shift $\varepsilon = 10^{-9}$), and after 300 iterations no mixing (1, 0.5, 0.2) has converged. The loop ends either in the $B_c = 1$-like state ($\nu_c = 1.9959$, single pole at -10.02, E higher than the direct minimum by 1.6×10^{-4} and 6.4×10^{-4}) or in a state with $R_c = 0$.

Maximum-overlap (MOM-like) selection. In each embedding solve, the eigenstate with the largest overlap with the previous Φ_I is taken instead of the lowest one (`follow=True`; degenerate subspaces are projected). One step from the direct-minimization state then gives a residual of 6.1×10^{-2} ($V/g = 0.707$) and 2.2×10^{-3} ($V/g = 1.414$), instead of 1.41, but it is still not a fixed point at $\varepsilon = 10^{-9}$. A filled ghost orbital has ${}^c\Delta$ eigenvalue 1.0000, so $S_c \sim \sqrt{\varepsilon}$ and λ^c, obtained from S_c^{-1}, is ill conditioned for small ε (this is my reading of the mechanism). The followed state is not in a dense part of the spectrum here: its nearest neighbours are 1.1 and 0.9 ($V/g = 0.707$), 2.0 and 0.14 ($V/g = 1.414$) away, apart from degenerate pairs at lower energy, which are projected. After that the loop drifts to the $B_c = 1$-like state (at $V/g = 1.414$, mixing 0.2 stays there, E higher by 6.4×10^{-4}, pole -10.01) or to the trivial product state ($E = -22$, $R_c = 0$).

Dependence on the shift (300 iterations, mixings 1.0, 0.5, 0.2, started from the direct-minimization state; "none" means no convergence to 10^{-10} in the residual):

shift ε	lowest-state selection	max-overlap, $V/g = 0.707$	max-overlap, $V/g = 1.414$
10^{-9}	none	none (ends in the $B_c = 1$-like or the product state)	none
10^{-8}	none	mixings 0.5, 0.2 converge to the product state ($R_c = 0$)	same
10^{-7}	none	none	mixings 0.5, 0.2 converge to the product state
10^{-6}	none	**mixings** 1.0**,** 0.2 **converge** (residual $\leq 3 \times 10^{-12}$) to a nearby minimum, E higher by 1.2×10^{-4}, pole -10.017	mixings 0.5, 0.2 converge to the product state
10^{-5}	none	none (stays at the $B_c = 1$-like state, residual 0.7–1.4)	**mixing** 0.2 **converges** (residual 6×10^{-13}) to a nearby minimum, E higher by 3.5×10^{-4}, pole -9.970
10^{-4}	none	none (residual 4×10^{-7}–2×10^{-6} after 300 iterations, E higher by 0.9–4×10^{-6})	none
10^{-3}	none	mixing 0.2 converges to the product state	none (product state)

The shift also changes the problem, so the direct-minimization energy used as reference changes with ε (at 10^{-4} and 10^{-3} it is -22.0412 and -22.0195 at $V/g = 0.707$). At the converged nearby minima ($\varepsilon = 10^{-6}$, $V/g = 0.707$, mixing 0.2, and $\varepsilon = 10^{-5}$, $V/g = 1.414$, mixing 0.2) restarting the direct minimizer from the loop point lowers E by 4×10^{-11} and 4×10^{-11} (by $\leq 10^{-14}$ for mixing 1.0 at $V/g = 0.707$): the loop point is itself a constrained local minimum, with the same single pole in G^h_{cc}.

Conclusion. The impurity loop reproduces a solution of this kind only with maximum-overlap state selection and a large regularization shift (10^{-6} at $\rho/\Omega = 0.5$, 10^{-5} at $\rho/\Omega = 1.0$; the other shifts give no convergence or the trivial product state); and then it finds a different, higher local minimum. The direct minimization needs no choice of eigenstate and is the more reliable solver.

Scope. The numbers are for $V/g = 0.707$ and 1.414, $t_c = 0.5$, $B_c = 3$ (`excited_check_golze.py` and `shift_sweep.py`; logs in `cederbaum_data/logs`). Which mixing converges depends on the starting point and on the one direct-minimization solution found (best of 5 random starts, 30 at $\varepsilon = 10^{-3}$, $V/g = 0.707$), so the table is a sample, not a map of all behaviour. The explanation in Sec. 4, points 3–4, is an interpretation of those numbers.